

Systematic Ablation Study — 10 Experiments

Method / Modification	Effect	Delta Pinball	Delta RMSE	Data Regime	Diagnosis
NWP Weather Features	[+++] Critical	+46.3%	Enables forecast	3.5K	NWP is the primary information source
Training Data 3.5K -> 28K	[+++] Critical	+61.0%	R2: -0.20 -> -0.09	3.5–28K	Data volume is the key bottleneck
LNN Gating over Pure SSM	[+] Marginal	+0.4%	18/24 horizons better	28K	Real but weak signal; needs more data
CRPS Auxiliary Loss (0.1x)	[X] Harmful	-5.1%	--	5K	Redundant with 99-Q pinball; distracts optimizer
Multi-Scale Conv Frontend	[X] Harmful	Overfit	Val loss diverges	3.5K	60K params too many for 3500 samples
Multi-Zone Joint Training	[X] Harmful	-67.8%	Catastrophic failure	35K	Per-zone scaler mismatch -> incomparable features
6x Strong Regularization	[X] Harmful	-40.7%	—	3.5K	Can't create information; underfitting dominates
30-Trial Random HP Search	[X] Ineffective	Best < v1	All 30 worse	3.5K	v1 config at exact optimal balance point
3-Model Ensemble + NWP Noise	[X] Harmful	-15%	—	3.5K	Ensemble needs diversity; noise degrades NWP signal
RevIN (Global, all variables)	[X] Harmful	-24~36%	—	3.5K	Destroys physical scale of NWP wind speed features

GEFCom2014 Zone 1 (3.5K) unless noted. [OK] = effective; [X] = degrades performance. Methods ordered by impact magnitude.